SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Solution Requirements

Date	23 June2026
Team ID	
Project Name	0pticrop:Smart Agriculture production optimization
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely log in using a username and password to access the application.	High
2	Authorization levels	Admin manages the system, researchers analyze data, and farmers can use crop recommendation features.	High

3	External interfaces	Connects with the machine learning model, crop dataset, and optionally weather or soil data sources.	Medium
4	Transactions processing	Accepts soil and environmental inputs, processes them through the ML model, and displays the recommended crop.	High
5	Reporting	Displays crop recommendation results and prediction summaries.	Medium
6	Business rules, etc.	Crop recommendation is based on Nitrogen, Phosphorous, Potassium, temperature, humidity, pH, and rainfall values.	High
7	Compliance to laws or regulations	Ensures secure handling of user data and follows applicable data privacy guidelines.	Medium
8	*Other*	Validates user inputs and provides error messages for invalid values.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Crop prediction should be generated quickly.	Response time < 2 seconds
2	Scalability	System should support multiple users simultaneously.	Supports 100+ concurrent users
3	Security & Data Privacy	User information should be securely stored and protected.	Secure login with encrypted communication (HTTPS)
4	Reliability & Availability	System should be available whenever users access it	99% uptime
5	Usability & Accessibility	Interface should be simple, responsive, and easy for farmers to use.	User-friendly interface with responsive design
6	*Other*	The application should be maintainable and easy to update in the future	Modular and maintainable code